

Homework 0 - The Basics

PHYS505 – Illinois Tech – Spring 2026

0.1 Coordinates

Describe the object/region that corresponds to each of the following ($\vec{x}$ is the position vector, $\vec{a}$ is a constant vector)

(a) in rectangular (Cartesian) coordinates (x, y, z)

- $x = 1$
- $x = y = 1$
- $x = y = z = 1$
- $x = y$
- $x = y = z$
- $x^2 + y^2 = 1$
- $\sqrt{x^2 + y^2} = z$
- $x^2 + y^2 = z^2$
- $\sqrt{x^2 + y^2} = |z|$

(b) in spherical coordinates (r, θ, ϕ)

- $r = 1$
- $\theta = 1$
- $\phi = 1$
- $r = \theta = 1$
- $r = \phi = 1$
- $\theta = \phi = 1$

(c) in cylindrical coordinates (ρ, ϕ, z)

- $\rho = 1$
- $\phi = 1$
- $z = 1$
- $\rho = \phi = 1$
- $\rho = z = 1$
- $\phi = z = 1$
- $\rho = \phi = z = 1$

(d) general

- $|\vec{x}| = 1$
- $\vec{a} \cdot \vec{x} = 0$
- $\vec{a} \cdot \vec{x} = |\vec{a}|$
- $\vec{a} \times \vec{x} = \vec{0}$
- $|\vec{a} \times \vec{x}| = |\vec{a}|$
- $\vec{a} \cdot \vec{x} = |\vec{a}|, |\vec{a} \times \vec{x}| \leq |\vec{a}|$

I am looking for simple answers (eg. “a plane parallel to the y - z plane with its x -intercept at 1”), not many pages of algebra. A sketch (if legible) would also be fine.

0.2 Vector calculus

Derive the following identities (mostly from the inside front cover of Jackson). $\vec{x}$ is the position vector, $\vec{a}$ is a constant vector, $r = |\vec{x}|$, $\hat{n} = \vec{x}/r$ is a unit radial vector, $f(r)$ is a well-behaved function of r .

$$\begin{array}{llll} \vec{\nabla} r = \hat{n} & \vec{\nabla} \cdot \vec{x} = 3 & \vec{\nabla} \times \vec{x} = \vec{0} & \nabla^2 r = \frac{2}{r} \\ & \vec{\nabla} \cdot \hat{n} = \frac{2}{r} & \vec{\nabla} \times \hat{n} = \vec{0} & \nabla^2 r^2 = 6 \\ \vec{\nabla} f(r) = \hat{n} \partial_r f & \vec{\nabla} \cdot [\hat{n} f(r)] = \frac{2}{r} f + \partial_r f & \vec{\nabla} \times [\hat{n} f(r)] = \vec{0} & \end{array}$$

Expressions for gradient, divergence and Laplacian are available on exam-I formula sheet (posted on Canvas) as well as the inside back cover of the book.